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Addressing the notion of trust around ChatGPT in the high-stakes use case of insurance

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ABSTRACT

The public discourse concerning the level of (dis)trust in ChatGPT and other applications based on large language models (LLMs) is loaded with generic, dread risk terms, while the heterogeneity of relevant theoretical concepts and empirical measurements of trust further impedes in-depth analysis. Thus, a more nuanced understanding of the factors driving the trust judgment call is essential to avoid unwarranted trust. In this commentary paper, we propose that addressing the notion of trust in consumer-facing LLM-based systems across the insurance industry can confer enhanced specificity to this debate. The concept and role of trust are germane to this particular setting due to the highly intangible nature of the product coupled with elevated levels of risk, complexity, and information asymmetry. Moreover, widespread use of LLMs in this sector is to be expected, given the vast array of text documents, particularly general policy conditions or claims protocols. Insurance as a practice is highly relevant to the welfare of citizens and has numerous spillover effects on wider public policy areas. We therefore argue that a domain-specific approach to good AI governance is essential to avoid negative externalities around financial inclusion. Indeed, as a constitutive element of trust, vulnerability is particularly challenging within this high-stakes set of transactions, with the adoption of LLMs adding to the socio-ethical risks. In light of this, our commentary provides a valuable baseline to support regulators and policymakers in unravelling the profound socioeconomic consequences that may arise from adopting consumer-facing LLMs in insurance.

1. A domain-specific approach confers greater specificity to the current dread risk discourse concerning ChatGPT and other large language models (LLMs)

The real-world impacts of deploying ChatGPT and other large language model (LLM)-based applications catapulted AI ethics and governance into mainstream media. In the past year, dramatic figures that shifted the public narrative to focus on the harms of deploying LLMs were cited in news headlines, including the 700,000 litres of freshwater consumed during the training of the GPT-3 model [1], the hourly wage of USD 2 for Kenyan data annotators creating hate speech datasets [2], and the billion-dollar copyright infringement lawsuit from the New York Times against OpenAI [3]. Turmoil over the proposed moratorium for LLMs [4] and the Sam Altman debacle [5] resonated with the latent safety concerns in the commercial race for AI leadership. A recent empirical study highlighted the contentious public sentiment towards ChatGPT and showed that the negative opinions regarding its use were based on ethical implications, including bias, manipulation, job displacement, and concerns about data privacy and security [6]. While the reflective enquiry in this area until now has largely been enmeshed in the normative target of making AI trustworthy, there is growing resistance to the overly simplistic rhetoric around trust used by government bodies [7]. The dread risk discourse evident in this space is especially problematic since it obfuscates consideration of the context in which the LLM is used and the precise task it is entrusted with. In fact, and as with language itself, the adoption of LLMs is constrained by domain-specific rules and structures (syntaxes), meanings (semantics), and situation-specific practices (pragmatics). We posit that embeddedness and situational dynamics are essential in the efforts to ensure the safe deployment of AI systems.

Examining LLM deployment in a specific domain with profound socioeconomic consequences is imperative for an in-depth discussion about trust and appropriate safeguards. In terms of high-risk scenarios, numerous publications have addressed areas such as LLMs in medicine

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and criminal justice systems. Insurance is another prime example of a consequential decision-making environment. As a practice, it is highly relevant to the welfare of citizens and has numerous spillover effects on wider public policy areas. As a sector, it already exercises considerable disciplinary power over populations: a power that can only increase with the widespread application of AI systems. The granularity of data reinforces the social control function of insurance by adjusting power relations between insurers and consumers. Accordingly, as a high-stakes use case, policymakers and researchers are presented with both a specific domain and a set of profound socioeconomic consequences that may arise from adopting LLMs.

Insurance regulators are keenly aware of the risks posed by AI deployment, as evidenced by the establishment of the European Insurance and Occupational Pensions Authority (EIOPA) Consultative Expert Group on Digital Ethics in Insurance in 2021 [8]. The expert group's deliberations centred on financial exclusion and unfair treatment of (vulnerable) consumers, including algorithmic bias, non-risk pricing, data misuse, and privacy concerns. For instance, US health insurer Cigna faces two class-action lawsuits for allegedly auto-denying medical expense claims in batches using its procedure-to-diagnosis (PXDX) system without reviewing patient files [9]. Cigna's medical directors reportedly denied over 300,000 claims in two months, taking an average of only 1.2 seconds to review each case [10]. The concerns regarding the PXDX system are among a growing number of incidents involving the ethical misuse of AI [9]. These real-world incidents underline the need to translate the high-level trustworthiness rhetoric into domain-specific practices, from which the following question arises: How is trust defined in the context of AI and LLMs and the insurance industry, and should regulators encourage such trust?

The insurance industry provides an excellent use case to address the notions of trust and trustworthiness in LLM-based systems in high-stakes decision scenarios. The concept of trust is germane to the insurance setting due to the product's highly intangible nature coupled with elevated levels of risk, complexity, and information asymmetry. Despite its well-established significance, little attention has been paid to the drivers of trust in insurance transactions [11]. Although the current state of trust research is still fluid, converging findings demonstrated two essential components for interpersonal trust relationships: a willingness to be vulnerable and a positive expectation concerning the other party's intentions and behaviour [12,13]. This consensus draws heavily on the definition by Mayer and colleagues [14] and the affiliated scale they developed [15]: a conceptual base that is gaining traction in the human-AI interaction field [16,17]. Vulnerability and risk are constitutive elements of trust, and their importance is significantly elevated in the domain of insurance. Task characteristics, interdependency, power (im)balances, and other contextual factors inevitably influence the perceived level of risk in the trust interaction with LLM-based systems.

The insurance sector plays a dual role in current discussions about the safe deployment of LLMs. As a user, insurance can harness LLMs to make unstructured data available and increase efficiency while recent advances in large multimodal models (LMMs) can result in another leap in possible use cases. Narrowing our focus to consumer-facing LLM-based applications in front-end processes, such as sales, customer service, and underwriting, allows us to adopt a more nuanced perspective. Additionally, insurance is strategically positioned to provide risk management support and transfer mechanisms to LLM developers. The sector's substantial risk management expertise can thus help shape the agenda and break the cycle of existential risk and emergency. Thus, in this commentary paper, we propose to add greater specificity to the current discourse around (dis)trust in ChatGPT and other LLM-based systems by moving beyond a high-level principle-based approach to a domain-specific view.

2. Insurance-specific considerations can ground the notion of trust in human-LLM interactions

Prior research stresses the importance of promoting appropriate trust levels in system capabilities to mitigate the misuse and disuse of automation [18]. We extend the interpersonal trust definition in Ref. [14] to a potential human-LLM trust relationship by exploring four key elements: (1) relationality, (2) vulnerability, (3) willingness, and (4) positive expectation.

2.1. Relationality

Trust can manifest within a relationship between two parties, such as an insurance consumer and an LLM-based application, through a particular action relevant to the trustor (i.e. the consumer trusts the LLM-based application to do a certain task). This interdependence introduces a fundamental element of risk and uncertainty and raises concerns about the impact of low levels of financial literacy – or digital skills in general – on interactions with insurance-related LLM-based systems. We propose that trust can generally extend to LLM-based systems because the trustors are willing to make themselves vulnerable by relying on the LLM's output under uncertainty, irrespective of its inherent capacity for volition or moral agency [18,19]. The fundamental elements of relationality and uncertainty remain the pivotal contextual conditions for trust development in LLMs.

2.2. Vulnerability

The trustors can make themselves vulnerable to an LLM system. Vulnerability is central to trust, denoting potential adversity or harm. As a consequence, trust becomes imperative in scenarios where outcomes are unpredictable or beyond the complete control of the trustor, particularly when the trustor does not (or cannot) exercise any monitoring or controlling mechanisms. Hence, saying 'I trust you', means 'I am willing to be vulnerable and to take a risk' – vulnerability and risk drive trust [20].

In insurance, the duty of utmost good faith requires the contracting parties to be honest and accurate and to disclose all material facts. A prominent risk of LLMs is that they are prone to generating plausible-sounding but factually incorrect information, especially when faced with unfamiliar customer queries. This hallucination risk is partly due to 'world knowledge gaps' [21] resulting from underreported information in the training data set or prompt engineering techniques. In this context, it is particularly problematic that – despite inherent inaccuracies – text produced in a convincingly high register may retain substantial rhetorical power. Significantly, there is no fixed relationship between the syntax quality and text accuracy that we are generally accustomed to and this is a grey area that raises significant concerns about potential deception. In Ref. [22], Rice and colleagues provide guidance for researchers on fact-checking the output to ensure the accuracy of information provided by ChatGPT. As a heavily regulated financial service, the sector must be alert to the risks of mis-selling insurance products.

Moreover, consumers may be inappropriately nudged to adopt particular behaviours or make purchases based on how an LLM response is phrased and which information is presented first. To process information and make decisions more quickly, consumers tend to consider only the most immediate information and have difficulty deviating from an initial value. These well-documented cognitive biases, such as 'What You See Is All There Is' (WYSIATI) and 'anchoring' [23] hamper adequate risk assessment and insurance purchase decisions. Since the output of an LLM varies based on how the input prompt is phrased, the model may work less efficiently for users deploying dialect, slang, or non-native terms. Such vulnerability is amplified by the use of complex insurance jargon which necessitates the use of a glossary to enable a layperson to fully comprehend the meaning of technical terms.

Insurance data scientists can thus customise pre-trained LLMs with domain-specific expertise. Combining Retrieval-Augmented Generation (RAG) and finetuning can ground query responses in information from insurer databases, similar resolved queries, specific terminology, and tone. Empathy is especially valued during the First Notice of Loss (FNOL) process which is the 'moment of truth' when it is revealed whether the insurer will keep their promise to provide coverage. However, the adaptation techniques greatly depend on data availability and quality, which poses a serious challenge to the insurance sector which is hampered by the legacy system dilemma. Moreover, real-world incidents of technical errors in other domains (e.g. the erroneously retrieved airfare information by Air Canada's LLM-based chatbot [9]) and malicious security attacks via LLM jailbreaking (e.g. the swearing DPD customer support LLM-based chatbot [9]) illustrate limitations. The provision of incorrect or harmful advice in insurance customer acquisition and retention is particularly alarming as it can lead to protection gaps or lock-in effects on consumers, thereby reducing their propensity to shop around at renewal.

2.3. Willingness

Trust, in essence, does not encompass risk but rather embodies the consumer's readiness to assume risk in a relationship with a specific LLM-based application. The consumer adopts a trusting attitude in anticipation of a risky situation, yet this attitude precedes any actual action [24]. Comparable to an intention to act, trust is rooted in particular beliefs and determines future behavioural responses.

Consumers may erroneously attribute human communicative intent and other characteristics, such as genuine concern, altruism, critical thinking, or a sense of justice, to LLMs. As such, they may overestimate the text quality and forego further reflection or exercise of control mechanisms, such as cross-checking with different prompts or other sources. In other words, we trained 'machines that can mindlessly generate text, [b]ut we haven't learned how to stop imagining the mind behind it' [25]. This could also lead to the over-sharing of personal information, which, in the context of insurance, could result in a sharp hike in premiums or the insurer declining to underwrite coverage. Privacy infringement refers to the potential risk of leaking highly sensitive information, and concerns regarding consumer autonomy for underwriting and pricing purposes and technical remedies, such as differential privacy, fall short of addressing this risk. Autonomy can range from choosing a preferred communication platform and informed consent to more fundamental surveillance issues, potentially affecting access to essential financial services vital for societal participation and welfare.

2.4. Positive expectation

The willingness to trust is based on a positive and confident expectation that the other party will keep their commitment. This anticipation is exemplified in one of the measure items stating that the trustee will 'keep [the trustor's] interests in mind when making decisions' [15]. Thus, trust, as a cognitive and deliberate choice, is partly anchored in specific beliefs that the LLM-based application is trustworthy. While insurance is often disparaged as one of the least trusted industries overall, closer scrutiny of specific relationships, such as broker-client engagements, reveals high levels of reciprocal trustworthiness. The same applies to ambivalent societal perceptions of AI systems. For instance, while a recent global study found that most people expressed fear and concern, and are generally wary of AI systems, trust levels are higher in specific use cases with a direct benefit [26].

European insurance legislation requires insurance professionals to act in the consumer's best interest (e.g. Insurance Distribution Directive (IDD)). LLMs trained on (non-representative or not-fit-for-purpose) datasets containing protected attributes or closely related proxy variables can create or magnify unfair discrimination and bias, thereby disadvantaging specific groups. LLMs can thus mirror the inherent ambiguities in the judgment of insurance professionals, such as false accusations of fraud. The operation of LLMs is also likely to create links between certain social phenomena and specific groups of people. In a domain where, by definition, perceived risk has an impact on prices, this could have profound consequences. In the short to medium term, bespoke application programming interfaces (APIs) will likely acquire LLMs' access to insurance databases which will arguably increase the potential for negative externalities, such as non-risk pricing and arbitrary underwriting decisions.

These initial considerations serve as a starting point to ground a sector-specific understanding of trust in LLMs.

3. Preliminary conclusions and future research directions

The storm of controversy provoked by the public release of ChatGPT has alerted policymakers to the issue of effectively regulating generative AI systems in the AI Act. This legislation, proposed by the European Commission (EC) in 2021 and approved by the European Council on 21 May 2024, imposes extensive risk management and transparency stipulations on deployers of high-risk AI systems, and it is anticipated that some insurance-specific AI systems will be designated as being high-risk. In the interim prior to statute enactment, the EC has published voluntary guidance in the form of the International Code of Conduct for Organizations Developing Advanced AI Systems [27]. Given its core competency in risk management, the insurance sector must play a key role in this development, and we propose that this area presents a unique opportunity for initiating a comprehensive and well-founded discussion on AI use in general, and LLM use in particular.

Context is essential for the development of appropriate governance regimes and we maintain that initiatives involved in deriving universal rules or codes of conduct for LLM use must, at the very minimum, be informed by specific use cases. While it is important to consider the broader societal impacts of ChatGPT beyond a single domain [28], situational dynamics and context critically influence public perception [6], trust levels, and the perceived sufficiency of current regulations and safeguards [26]. Working through the issue in high-risk domains such as insurance can offer some direction for policymakers seeking to protect the interests of European citizens. The use of LLMs by insurance companies poses several risks for consumers, including financial exclusion, mis-selling, and information asymmetry. Bespoke LLMs with direct access to insurance databases will only increase the potential for such negative externalities. Moreover, axiomatic to insurance business practice is a future promise to meet all contractual commitments. The addition of this temporal dimension to the current debate is significant as it further increases the vulnerability tariff, thereby redoubling the potential cost of trusting the LLM-generated text. Both the extended temporal component and the *trust not to* question (LLM safeguards) in insurance underline the need for vigilance and, potentially, the exercise of the precautionary principle.

Insurance business processes harnessing the power of AI systems may signal a paradigm shift in insurance. The origins of insurance practice lay in sharing risk and possessing partial information on insured citizens. To take John Rawls as a starting point and extend the metaphor, a net curtain has obscured future elements [29]. Hence, 'Fortuna', or luck, was an accepted aspect of the early insurance practitioner landscape [30]. However, something approaching omniscience concerning consumer data is possible in a world of the Internet of Things, wearables, rapid advances in gathering, sequencing, and storing genetic information, social media data, and novel AI-based techniques. This is especially relevant for insurance products with elevated perceived risk and complexity levels, such as life and health insurance. While the nature of debates on the social impact of LLMs is somewhat amorphous and lacking in precision, we propose that examining the impact of this technology in insurance, especially in life and health lines, offers real traction to reinterpret the notion of trust(worthiness) in AI systems.

CRediT authorship contribution statement

Juliane Ressel: Writing – review & editing, Writing – original draft, Conceptualization. **Michaele Völler:** Writing – review & editing, Supervision, Conceptualization. **Finbarr Murphy:** Writing – review & editing, Supervision. **Martin Mullins:** Writing – review & editing, Writing – original draft, Conceptualization.

Declaration of competing interest

The authors declare no competing interests.

Data availability

No data was used for the research described in the article.

References

[1] P. Li, J. Yang, M.A. Islam, S. Ren, Making AI less "thirsty": uncovering and addressing the secret water footprint of AI models, ArXiv (2023), https://doi.org/10.48550/arXiv.2304.03271 abs/2304.03271.

[2] B. Perrigo, Exclusive: OpenAI Used Kenyan Workers on Less than $2 Per Hour to Make ChatGPT Less Toxic, TIME, 2023. URL: https://time.com/6247678/openai-chatgpt-kenya-workers/. (Accessed 19 March 2024).

[3] M.M. Grynbaum, R. Mac, The Times sues OpenAI and microsoft over A.I. Use of copyrighted work, N. Y. Times (27 December 2023). URL: https://www.nytimes.com/2023/12/27/business/media/new-york-times-open-ai-microsoft-lawsuit.html. (Accessed 19 March 2024).

[4] Future of Life Institute, Policymaking in the Pause. What can policymakers do *now* to combat risks from advanced AI systems?, URL: https://futureoflife.org/document/policymaking-in-the-pause/, 2023. (Accessed 19 March 2024).

[5] N. Jones, What the OpenAI drama means for AI progress – and safety, Nature 623 (23 November 2023) 898–899, https://doi.org/10.1038/d41586-023-03700-4.

[6] Y. Xing, J.Z. Zhang, G. Teng, X. Zhou, Voices in the digital storm: unraveling online polarization with ChatGPT, Technol. Soc. 77 (2024) 102534, https://doi.org/10.1016/j.techsoc.2024.102534.

[7] J. Laux, S. Wachter, B. Mittelstadt, Trustworthy artificial intelligence and the European Union AI act: on the conflation of trustworthiness and acceptability of risk, Regulation & Governance 18 (2023) 3–32, https://doi.org/10.1111/rego.12512.

[8] European Insurance and Occupational Pensions Authority (EIOPA), Artificial Intelligence Governance Principles, towards Ethical and Trustworthy Artificial Intelligence in the European Insurance Sector – A Report from EIOPA's Consultative Expert Group on Digital Ethics in Insurance, Publications Office of the European Union, 2021. https://data.europa.eu/doi/10.2854/49874.

[9] AI, Algorithmic, and Automation Incidents and Controversies (AIAAIC) Repository (2024). URL: https://www.aiaaic.org/aiaaic-repository. (Accessed 19 March 2024).

[10] P. Rucker, M. Miller, D. Armstrong, How Cigna saves millions by having its doctors reject claims without reading them, The Capitol Forum and ProPublica (25 March 2023). URL: https://www.propublica.org/article/cigna-pxdx-medical-health-insurance-rejection-claims. (Accessed 19 March 2024).

[11] C. Courbage, C. Nicolas, Trust in insurance: the importance of experiences, J. Risk Insur. 88 (2020) 263–291, https://doi.org/10.1111/jori.12324.

[12] M.D. Baer, J.A. Colquitt, Why do people trust? Moving toward a more comprehensive consideration of the antecedents of trust, in: R.H. Searle, A. M. Nienaber, S.B. Sitkin (Eds.), Routledge Companion to Trust, Routledge, London and New York, 2018, pp. 163–182.

[13] D.M. Rousseau, S.B. Sitkin, R.S. Burt, C. Camerer, Not so different after all: a cross-discipline view of trust, Acad. Manag. Rev. 23 (3) (1998) 393–404, https://doi.org/10.5465/AMR.1998.926617.

[14] R.C. Mayer, J.H. Davis, F.D. Schoorman, An integrative model of organizational trust, Acad. Manag. Rev. 20 (3) (1995) 709–734, https://doi.org/10.2307/258792.

[15] R.C. Mayer, J.H. Davis, The effect of the performance appraisal system on trust for management: a field quasi-experiment, J. Appl. Psychol. 84 (1) (1999) 123–136, https://doi.org/10.1037/0021-9010.84.1.123.

[16] T.A. Bach, A. Khan, H. Hallock, G. Beltrão, S. Sousa, A systematic literature review of user trust in AI-enabled systems: an HCI perspective, Int. J. Hum. Comput. Interact. 40 (5) (2022) 1251–1266, https://doi.org/10.1080/10447318.2022.2138826.

[17] O. Vereschak, G. Bailly, B. Caramiaux, How to evaluate trust in AI-assisted decision making? A survey of empirical methodologies, Proc. ACM Hum.-Comput. Interact. 5 (CSCW2) (2021), https://doi.org/10.1145/3476068. Art. 327.

[18] J.D. Lee, K.A. See, Trust in automation: designing for appropriate reliance, Hum. Factors 46 (1) (2004) 50–80, https://doi.org/10.1518/hfes.46.1.50_30392.

[19] D.H. McKnight, M. Carter, J.B. Thatcher, P.F. Clay, Trust in a specific technology: an investigation of its components and measures, ACM Trans. Manage. Inf. Syst. 2 (2011), https://doi.org/10.1145/1985347.1985353. Article 12.

[20] J. Davis, Building Trust. TEDx Talk. TEDxUSU, Utah State University), 2014. URL: https://youtu.be/s9FBK4eprmA?si=uzoLDqNXoSDqk2Vz. (Accessed 19 March 2024).

[21] K. Mahowald, A.A. Ivanova, I.A. Blank, N.G. Kanwisher, J.B. Tenenbaum, E. Fedorenko, Dissociating language and thought in large language models: a cognitive perspective, ArXiv (2023), https://doi.org/10.48550/arXiv.2301.06627 abs/2301.06627.

[22] S. Rice, S.R. Crouse, S.R. Winter, C. Rice, The advantages and limitations of using ChatGPT to enhance technological research, Technol. Soc. 76 (2024) 102426, https://doi.org/10.1016/j.techsoc.2023.102426.

[23] D. Kahneman, Thinking, Fast and Slow, Farrar, Straus and Giroux, New York, 2011.

[24] F.D. Schoorman, M.M. Wood, C. Breuer, Would trust by any other name smell as sweet? Reflections on the meanings and uses of trust across disciplines and context, in: B.H. Bornstein, A.J. Tomkins (Eds.), Motivating Cooperation and Compliance with Authority: the Role of Institutional Trust, Nebraska Symposium on Motivation, Springer International Publishing, Cham, 2015, pp. 13–35, https://doi.org/10.1007/978-3-319-16151-8_2.

[25] E. Bender, E. Weil, You Are Not a Parrot. And a chatbot is not a human. And a linguist named Emily M. Bender is very worried what will happen when we forget this, Intelligencer, New York Magazine (2023). URL: https://nymag.com/intelligencer/article/ai-artificial-intelligence-chatbots-emily-m-bender.html. (Accessed 19 March 2024).

[26] N. Gillespie, S. Lockey, C. Curtis, J. Pool, A. Akbari, Trust in Artificial Intelligence: A Global Study, The University of Queensland; KPMG Australia, Brisbane, Australia; New York, United States, 2023, https://doi.org/10.14264/00d3c94.

[27] European Commission, Hiroshima process international code of conduct for advanced AI systems, URL: https://digital-strategy.ec.europa.eu/en/library/hiroshima-process-international-code-conduct-advanced-ai-systems, 30 October 2023. (Accessed 19 March 2024).

[28] X. Wei, X. Chu, J. Geng, Y. Wang, P. Wang, H. Wang, C. Wang, L. Lei, Societal impacts of chatbot and mitigation strategies for negative impacts: a large-scale qualitative survey of ChatGPT users, Technol. Soc. 77 (2024) 102566, https://doi.org/10.1016/j.techsoc.2024.102566.

[29] J. Rawls, A Theory of Justice, Harvard Univ. Press, 1971. URL: https://www.hup.harvard.edu/books/9780674000780.

[30] G.W. Clark, Betting on Lives: the Culture of Life Insurance in England, 1695-1775, Manchester Univ. Press, 1999.